COMS 311: Homework 2
Due: Feb 23, 11:59pm
Total Points: 50

Late submission policy. Any assignment submission that is late by not more than two business days from the deadline will be accepted with 20% penalty for each business day. That is, if a homework is due on Friday at 11:59 PM, then a Monday submission gets 20% penalty and a Tuesday submission gets another 20% penalty. After Tuesday no late submissions are accepted.

Submission format. Homework solutions will have to be typed. You can use word, LaTeX, or any other type-setting tool to type your solution. Your submission file should be in pdf format. Do **NOT** submit a photocopy of handwritten homework except for diagrams that can be hand-drawn and scanned. We reserve the right **NOT** to grade homework that does not follow the formatting requirements. Name your submission file: `<Your-net-id>-311-hw2.pdf`. For instance, if your netid is `asterix`, then your submission file will be named `asterix-311-hw2.pdf`. Each student must hand in their own assignment. If you discussed the homework or solutions with others, a list of collaborators must be included with each submission. Each of the collaborators has to write the solutions in their own words (copies are not allowed).

General Requirements

- When proofs are required, do your best to make them both clear and rigorous. Even when proofs are not required, you should justify your answers and explain your work.
- When asked to present a construction, you should show the correctness of the construction.

Some Useful (in)equalities

- $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$
- $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$
- $2^{\log_2 n} = n$, $a^{\log_b n} = n^{\log_b a}$, $n^{n/2} \le n! \le n^n$, $\log x^a = a \log x$
- $\log(a \times b) = \log a + \log b$, $\log(a/b) = \log a - \log b$
- $a + ar + ar^2 + ... + ar^{n-1} = \frac{a(r^n - 1)}{r-1}$
- $1 + \frac{1}{2} + \frac{1}{2^2} + ... + \frac{1}{2^n} = 2(1 - \frac{1}{2^{n+1}})$
- $1 + 2 + 4 + ... + 2^n = 2^{n+1} - 1$

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1. (**10 pts**) Using a table size of 11, and a hash function $h(x) = 3x + 2$ for positive integers, draw the resulting hash table array when the integers

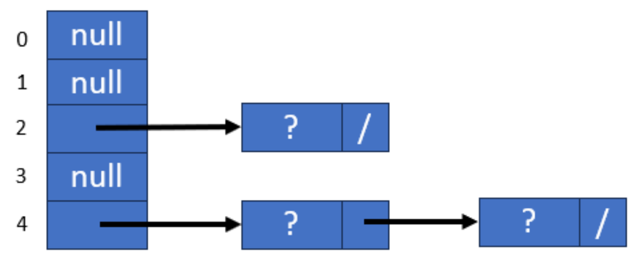
$$\mathbf{1, 17, 10, 15, 14, 43, 4, 21, 23, 24, 19, 41, 42, 9, 44}$$

are added to an empty hash table in this order. Note that for this hash function, x is stored in the table at index $h(x)\ \%\ n$, where n is the table size.

   (a) Using a chaining hash table that is not enlarged when elements are added.

   (b) Using a chaining hash table that is doubled in size when the table reaches a load factor of $\alpha > 0.75$.

   An example drawing of a hash table of table size 5 with 3 elements is shown below.



2. (**10 pts**) Consider the following recurrence relation:

$$T(n) = \begin{cases} 1 & \text{if } n = 1 \\ 7 \cdot T(\frac{n}{2}) + 1 & \text{if } n > 1 \end{cases}$$

   (a) Use the Master theorem to show that $T(n) \in \Theta(n^{\log_2(7)})$.

   (b) Use induction to prove that $T(n) = \frac{1}{6}(7n^{\log_2(7)} - 1)$.

3. (**10 pts**) Without using the Master Theorem, give the asymptotic upper bounds on the following recurrence relations. Note that methods found in chapter 4 of the text may be useful.

(a) $T(n) = \begin{cases} 1 & \text{if } n < 2 \\ 4 \cdot T(\frac{n}{2}) + n & \text{if } n \geq 2 \end{cases}$

(b) $T(n) = \begin{cases} 1 & \text{if } n < 2 \\ 2 \cdot T(\frac{n}{2}) + n \log n & \text{if } n \geq 2 \end{cases}$

(c) $T(n) = \begin{cases} 1 & \text{if } n < 3 \\ T(\frac{n}{3}) + n & \text{if } n \geq 3 \end{cases}$

(d) $T(n) = \begin{cases} 1 & \text{if } n < 3 \\ 9 \cdot T(\frac{n}{3}) + n^{2.5} & \text{if } n \geq 3 \end{cases}$

(e) $T(n) = \begin{cases} 1 & \text{if } n < 3 \\ T(n-2) + n^2 & \text{if } n \geq 3 \end{cases}$

4. (**10 pts**) If possible, use the Master theorem to give bounds on the following recurrence relations.

   (a) $T(n) = \begin{cases} 1 & \text{if } n < 2 \\ 4 \cdot T(\frac{n}{2}) + n & \text{if } n \geq 2 \end{cases}$

   (b) $T(n) = \begin{cases} 1 & \text{if } n < 2 \\ 2 \cdot T(\frac{n}{2}) + n \log n & \text{if } n \geq 2 \end{cases}$

   (c) $T(n) = \begin{cases} 1 & \text{if } n < 3 \\ 3 \cdot T(\frac{n}{3}) + n & \text{if } n \geq 3 \end{cases}$

   (d) $T(n) = \begin{cases} 1 & \text{if } n < 2 \\ 2 \cdot T(\frac{n}{4}) + \sqrt{n} & \text{if } n \geq 2 \end{cases}$

5. (**10 pts**) The algorithm below computes nothing useful. It takes as a parameter an array A of integers. Note that $A.length$ returns the length of the array, and $A.sub(s, l)$ returns a new array (with elements copied) of length l with values copied from A starting at index s.

Algorithm 1 Wacky(A)

if $A.length == 1$ **then**
    $A[0]$ += 1;
else
    int $m = \lfloor A.length/2 \rfloor$;
    $Wacky\ (A.sub(0, m))$;
    $Wacky\ (A.sub(m, m))$;
    $Wacky\ (A.sub(0, 1))$;

(a) Write a recurrence relation that gives the running time of **Wacky**.

(b) Use the Master theorem to give a bound on the running time in terms of n.